

Sirui Liang

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Major: Artificial Intelligence

GitHub: <https://github.com/Astra-Ranger>



Education

Great Bay University | Undergraduate Student | Artificial Intelligence

Sep 2025 - Present

GPA: 3.70 / 4.00 (Top 15%)

Key Courses: Mathematical Analysis, Linear Algebra, Introduction to Computer Science

Projects & Competitions

Virtual Photography System Based on Spatiotemporal Matting and Dynamic Relighting

Mar 2026 - Present

Project Lead | Advisor: Xiaodong Cun, Assistant Professor at GBU

National-level Innovation and Entrepreneurship Training Program for Undergraduates

Deep Learning / PyTorch / SAM2 / Video Matting / Multi-GPU Training

- Developed a SAM2-driven spatiotemporal matting and dynamic relighting system to address temporal flickering, boundary-detail loss, and lighting inconsistency in virtual photography.
- Independently reviewed 20+ CCF-A papers, reproduced video matting baselines(MatAnyone), trained VM240K and related data on 8 RTX 4090 GPUs, and explored skip-connection improvements for encoder detail loss.

CS101 Copilot (Intelligent Learning Assistant for Introduction to Computer Science)

Apr 2026 - May 2026

Project Lead | RAG Learning Assistant | Flask / ChromaDB / LangChain / JavaScript

GitHub: <https://github.com/Astra-Ranger/CS101-Copilot>

- Built a course-grounded RAG assistant for CS101, addressing the limitations of general LLMs in accurately answering domain-specific queries.
- Led 10 core features, including slide browsing, ChromaDB retrieval, page-level citation tracing, SSE streaming status, notebooks, and AI autocomplete.

Campus Cafe Ordering Platform

Feb 2026

Independent Project | Full-Stack Development and Server Deployment | FastAPI / SQLite / JavaScript

- Built a customer ordering and staff management platform to improve offline ordering efficiency and centralize menu and order operations for a campus cafe.
- Independently delivered the FastAPI backend, SQLite data layer, frontend pages, permission control, and server deployment; the platform has supported 30+ concurrent ordering users at peak.

AI Plan Table for HarmonyOS

Apr 2026 - May 2026

Project Lead | 2026 HarmonyOS Innovation Race(Results Pending) | HarmonyOS / ArkTS / ArkUI

GitHub: <https://github.com/Astra-Ranger/HarmonyPlanTable>

- Designed a HarmonyOS Eisenhower Matrix task planner to address the limitation of ordinary reminder apps that only list events without prioritization or contextual suggestions.
- Implemented 5 core capabilities, including the ArkTS / ArkUI task board, Preferences persistence, deadline reminders, AI task parsing, and AI recommendations.

Skills

Programming	Proficient in Python (PyTorch, NumPy) and JavaScript; familiar with HTML / CSS development
Algorithms	Familiar with Transformer and ViT architectures; experienced in video matting and instance segmentation; experienced in model training, metric evaluation, and RAG application development
Engineering	Familiar with Linux commands; proficient in Git; able to set up server cluster environments and troubleshoot dependency conflicts, and training configurations
AI Collaboration	Familiar with Vibe Coding workflows; proficient in using Claude Code, Codex, and other AI agents for requirement breakdown, code generation, debugging, and Skill workflow development
General	CET-4(510); able to read English papers and technical documentation; proficient in VS Code, Jupyter Notebook, and Photoshop

Summary

Artificial Intelligence undergraduate focused on computer vision and LLM application engineering, with experience in video matting experiments, RAG system development, and full-stack project delivery. Capable of literature review, code reproduction, model training, server deployment, and project delivery; accustomed to driving iteration through experiment records and measurable results.